MASAREN Platform

AI Services Financial Report

Cost Analysis, Budget Planning & Audit Documentation

Egyptian Pound (EGP) Budget Range: 500–10,000/month

MASAREN Financial Planning Division

August 2026

Core Principle: All infrastructure costs are borne by the platform operator. This report covers only cloud-based AI inference services.

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1 Executive Summary

This financial report provides a comprehensive cost analysis of Artificial Intelligence (AI) services deployed on the MASAREN Impact Infrastructure as a Service (IIaaS) platform. The analysis is scoped exclusively to cloud-based AI model inference costs—the platform operator provides all other infrastructure (servers, databases, networking, storage) at no incremental AI-related cost.

1.1 Key Financial Metrics

Metric	Value
Monthly Budget Range	500–10,000 EGP per AI service
Exchange Rate	1 USD = 50.5 EGP (August 2026)
Total Monthly AI Cost (Conservative)	1,219 EGP (USD 24.14)
Total Monthly AI Cost (Production)	8,235 EGP (USD 163.07)
Total Monthly AI Cost (Enterprise)	10,707 EGP (USD 212.02)
Annual AI Cost Range	14,628–128,484 EGP
AI Cost as % of Total Platform Budget	100% (AI-only report)
Number of AI Services	7 distinct services
Number of AI Models	6 models across 3 providers
Cost Recovery Model	User-charged (100% recovery target)

Table 1: Executive Summary Metrics

1.2 Scope of This Report

This report covers the following AI services that operate within the MASAREN platform:

1. **Vision AI Service** – Image/video evidence analysis and verification
2. **Fraud Detection Service** – Geospatial and behavioral fraud scoring
3. **Impact Economist Agent** – Social value computation and SDG scoring
4. **Graph Synthesis Service** – Knowledge graph construction and PMS delta analysis
5. **Chat Assistant Service** – User-facing conversational AI support
6. **Verification Orchestrator** – Multi-agent CrewAI verification pipeline
7. **Report Generator Service** – Narrative synthesis for impact reports

Out of scope: Server hosting, database operations, Kafka message broker, Redis cache, Qdrant vector store, CDN, domain/DNS, SSL certificates, monitoring (Prometheus/Grafana), and all non-AI application logic.

2 AI Platform Pricing Reference (August 2026)

All pricing is sourced from official provider documentation as of August 2026. Prices are per 1 million tokens (MTok) in USD.

2.1 OpenAI Pricing

Model	Input $MTok	Cached $MTok	Output $MTok	Context
GPT-4o	2.50	1.25	10.00	128K
GPT-4o mini	0.15	0.075	0.60	128K
GPT-4.1	2.00	0.50	8.00	1M
GPT-4.1 mini	0.40	0.10	1.60	1M
text-embedding-3-small	0.02	—	—	8K

Table 2: OpenAI Pricing (source: developers.openai.com/api/docs/pricing)

2.2 Google Gemini Pricing

Model	Input $MTok	Cached $MTok	Output $MTok	Context
Gemini 1.5 Flash	0.075	0.01875	0.30	1M
Gemini 1.5 Pro	1.25	0.3125	5.00	2M
Gemini 2.0 Flash	0.10	—	0.40	1M
text-embedding-004	0.10	—	—	2M

Table 3: Google Gemini Pricing (source: ai.google.dev/gemini-api/docs/pricing)

2.3 OpenRouter / Gemma Pricing

Model	Input $MTok	Output $MTok	Context	License
Gemma 4 31B IT	0.12	0.36	256K	Apache 2.0

Table 4: OpenRouter Gemma Pricing (source: openrouter.ai)

2.4 Provider Cost Comparison

Capability	OpenAI	Google	OpenRouter	Cheapest
Flagship Text	GPT-4o: $2.50	Gemini 1.5 Pro: $1.25	—	Google
Budget Text	GPT-4o mini: $0.15	Gemini 1.5 Flash: $0.075	—	Google
Vision/Multimodal	GPT-4o: $2.50	Gemini 1.5 Pro: $1.25	—	Google
Chat/Conversation	GPT-4o mini: $0.15	Gemma 4 31B: $0.12	Gemma 4 31B: $0.12	Gemma
Embeddings	text-3-small: $0.02	embedding-004: $0.10	—	OpenAI

Table 5: Provider Cost Comparison per Capability

3 Backend AI Service Architecture

The MASAREN platform deploys 13 backend microservices. Of these, 7 are AI-dependent services that require cloud LLM inference. The remaining 6 services handle infrastructure concerns (Kafka message bus, vector store embeddings) that are covered within the AI budget.

3.1 AI Service Inventory

Service	AI Model(s)	Provider	Deployment
Vision AI Service	Gemini 1.5 Pro (vision)	Google	Kafka consumer
Fraud Detection Service	Gemini 1.5 Flash	Google	HTTP API + Kafka
Impact Economist Agent	GPT-4o	OpenAI	Kafka consumer
Graph Synthesis (Auditor)	GPT-4o	OpenAI	Kafka consumer
Chat Assistant	Gemma 4 31B IT	OpenRouter	HTTP API
Verification Orchestrator	Gemma 4 31B IT	OpenRouter	CrewAI agents
Report Generator	Gemini 1.5 Flash	Google	HTTP API

Table 6: AI Service to Model Mapping

3.2 Model Configuration Parameters

Each AI model is configured with specific parameters that affect cost:

Service	Temperature	Max Tokens	JSON Mode
Vision AI	0.2	2,048	No
Fraud Detection	0.2	2,048	Yes
Impact Economist	0.1	2,048	Yes
Graph Synthesis	0.1	2,048	Yes
Chat Assistant	0.3	2,048	No
Verification Orchestrator	0.1	2,048	No
Report Generator	0.2	2,048	No

Table 7: AI Model Configuration Parameters

3.3 AI Agent Event Pipeline

The AI services operate as an event-driven pipeline via Apache Kafka:

1. **EvidenceUploaded** triggers Vision AI and Fraud Detection in parallel
2. **EvidenceVerified** triggers Impact Economist Agent
3. **ImpactAssetCreated** triggers Graph Synthesis and Broker Agent
4. **VerificationRequested** triggers Verification Orchestrator (CrewAI)
5. **ReportGenerated** triggers Report Generator for narrative synthesis

Each event triggers exactly one LLM inference call per consumer, enabling precise cost-per-event calculation.

4 AI Services Cost Breakdown (500–10,000 EGP/month)

All costs are calculated using actual backend model configurations, real August 2026 provider pricing, and projected daily usage volumes. Exchange rate: 1 USD = 50.5 EGP.

4.1 Vision AI Services

Evidence image analysis using multimodal LLMs. The Vision AI Service processes user-submitted evidence photos and videos for SDG impact verification.

Provider	Model	Input Tok/day	Output Tok/day	Monthly EGP	Notes
Google	Gemini 1.5 Pro	150,000	50,000	2,923	Primary (vision)
OpenAI	GPT-4o	150,000	50,000	5,471	Fallback

Table 8: Vision AI Service – Monthly Cost (500–10,000 EGP range)

Calculation (Google Gemini 1.5 Pro):

$$\text{Monthly Input Cost} = \frac{150{,}000 \times 30}{1{,}000{,}000} \times \$1.25 = \$5.625$$

$$\text{Monthly Output Cost} = \frac{50{,}000 \times 30}{1{,}000{,}000} \times \$5.00 = \$7.50$$

$$\text{Total Monthly (USD)} = \$13.125$$

$$\text{Total Monthly (EGP)} = \$13.125 \times 50.5 = \textbf{663 EGP}$$

Calculation (OpenAI GPT-4o fallback):

$$\text{Monthly Input Cost} = \frac{150{,}000 \times 30}{1{,}000{,}000} \times \$2.50 = \$11.25$$

$$\text{Monthly Output Cost} = \frac{50{,}000 \times 30}{1{,}000{,}000} \times \$10.00 = \$15.00$$

$$\text{Total Monthly (USD)} = \$26.25$$

$$\text{Total Monthly (EGP)} = \$26.25 \times 50.5 = \textbf{1,326 EGP}$$

Recommended Configuration: Google Gemini 1.5 Pro at 663 EGP/month (within budget).

4.2 Fraud Detection AI

Geospatial and behavioral fraud scoring. Analyzes GPS coordinates, checksums, reference points, and content patterns to detect fraudulent evidence submissions.

Provider	Model	Input Tok/day	Output Tok/day	Monthly EGP	Notes
Google	Gemini 1.5 Flash	200,000	100,000	1,568	Primary (fast)
OpenAI	GPT-4o	200,000	100,000	4,054	Fallback

Table 9: Fraud Detection AI – Monthly Cost (500–10,000 EGP range)

Calculation (Google Gemini 1.5 Flash):

$$\text{Monthly Input Cost} = \frac{200{,}000 \times 30}{1{,}000{,}000} \times \$0.075 = \$0.45$$
$$\text{Monthly Output Cost} = \frac{100{,}000 \times 30}{1{,}000{,}000} \times \$0.30 = \$0.90$$
$$\text{Total Monthly (USD)} = \$1.35$$
$$\text{Total Monthly (EGP)} = \$1.35 \times 50.5 = \textbf{68 EGP}$$

Calculation (OpenAI GPT-4o fallback):

$$\text{Monthly Input Cost} = \frac{200{,}000 \times 30}{1{,}000{,}000} \times \$2.50 = \$15.00$$
$$\text{Monthly Output Cost} = \frac{100{,}000 \times 30}{1{,}000{,}000} \times \$10.00 = \$30.00$$
$$\text{Total Monthly (USD)} = \$45.00$$
$$\text{Total Monthly (EGP)} = \$45.00 \times 50.5 = \textbf{2,273 EGP}$$

Recommended Configuration: Google Gemini 1.5 Flash at 68 EGP/month (within budget).

4.3 Impact Economist AI

Social value computation and SDG alignment scoring. Calculates impact value in USD terms and maps evidence to UN Sustainable Development Goal indicators.

Provider	Model	Input Tok/day	Output Tok/day	Monthly EGP	Notes
OpenAI	GPT-4o	100,000	80,000	2,027	Primary (reasoning)
Google	Gemini 1.5 Pro	100,000	80,000	2,176	Fallback

Table 10: Impact Economist AI – Monthly Cost (500–10,000 EGP range)

Calculation (OpenAI GPT-4o):

$$\text{Monthly Input Cost} = \frac{100{,}000 \times 30}{1{,}000{,}000} \times \$2.50 = \$7.50$$
$$\text{Monthly Output Cost} = \frac{80{,}000 \times 30}{1{,}000{,}000} \times \$10.00 = \$24.00$$
$$\text{Total Monthly (USD)} = \$31.50$$
$$\text{Total Monthly (EGP)} = \$31.50 \times 50.5 = \textbf{1,591 EGP}$$

Recommended Configuration: OpenAI GPT-4o at 1,591 EGP/month (within budget).

4.4 Graph Synthesis AI

Knowledge graph construction and PMS (Platform Metric System) delta analysis. Auditor agent analyzes impact dimension changes and feeds into Neo4j knowledge graph.

Provider	Model	Input Tok/day	Output Tok/day	Monthly EGP	Notes
OpenAI	GPT-4o	50,000	40,000	1,014	Primary (structured)
Google	Gemini 1.5 Pro	50,000	40,000	1,088	Fallback

Table 11: Graph Synthesis AI – Monthly Cost (500–10,000 EGP range)

Calculation (OpenAI GPT-4o):

$$\text{Monthly Input Cost} = \frac{50{,}000 \times 30}{1{,}000{,}000} \times \$2.50 = \$3.75$$

$$\text{Monthly Output Cost} = \frac{40{,}000 \times 30}{1{,}000{,}000} \times \$10.00 = \$12.00$$

$$\text{Total Monthly (USD)} = \$15.75$$

$$\text{Total Monthly (EGP)} = \$15.75 \times 50.5 = \mathbf{795\ EGP}$$

Recommended Configuration: OpenAI GPT-4o at 795 EGP/month (within budget).

4.5 Chat Assistant AI

User-facing conversational AI for platform navigation, task guidance, and impact portfolio inquiries. Uses Gemma 4 31B IT via OpenRouter.

Provider	Model	Input Tok/day	Output Tok/day	Monthly EGP	No
OpenRouter	Gemma 4 31B IT	80,000	60,000	284	Primary (Apache

Table 12: Chat Assistant AI – Monthly Cost

Calculation (OpenRouter Gemma 4 31B IT):

$$\text{Monthly Input Cost} = \frac{80{,}000 \times 30}{1{,}000{,}000} \times \$0.12 = \$0.288$$

$$\text{Monthly Output Cost} = \frac{60{,}000 \times 30}{1{,}000{,}000} \times \$0.36 = \$0.648$$

$$\text{Total Monthly (USD)} = \$0.936$$

$$\text{Total Monthly (EGP)} = \$0.936 \times 50.5 = \mathbf{47\ EGP}$$

Recommended Configuration: OpenRouter Gemma 4 31B IT at 47 EGP/month (within budget).

4.6 Verification Orchestrator AI

Multi-agent CrewAI pipeline for complex evidence verification. Orchestrates multiple AI agents with debate/reflection pattern.

Provider	Model	Input Tok/day	Output Tok/day	Monthly EGP	Not
OpenRouter	Gemma 4 31B IT	120,000	100,000	426	CrewAI multi-age

Table 13: Verification Orchestrator AI – Monthly Cost

Calculation (OpenRouter Gemma 4 31B IT):

$$\text{Monthly Input Cost} = \frac{120{,}000 \times 30}{1{,}000{,}000} \times \$0.12 = \$0.432$$

$$\text{Monthly Output Cost} = \frac{100{,}000 \times 30}{1{,}000{,}000} \times \$0.36 = \$1.08$$

$$\text{Total Monthly (USD)} = \$1.512$$

$$\text{Total Monthly (EGP)} = \$1.512 \times 50.5 = \textbf{76 EGP}$$

Recommended Configuration: OpenRouter Gemma 4 31B IT at 76 EGP/month (within budget).

4.7 Report Generator AI

Narrative synthesis for COMPREHENSIVE-tier impact reports. Generates executive summaries, SDG indicator narratives, and financial analysis text.

Provider	Model	Input Tok/day	Output Tok/day	Monthly EGP	Notes
Google	Gemini 1.5 Flash	40,000	60,000	110	Primary (narrative)

Table 14: Report Generator AI – Monthly Cost

Calculation (Google Gemini 1.5 Flash):

$$\text{Monthly Input Cost} = \frac{40{,}000 \times 30}{1{,}000{,}000} \times \$0.075 = \$0.09$$

$$\text{Monthly Output Cost} = \frac{60{,}000 \times 30}{1{,}000{,}000} \times \$0.30 = \$0.54$$

$$\text{Total Monthly (USD)} = \$0.63$$

$$\text{Total Monthly (EGP)} = \$0.63 \times 50.5 = \textbf{32 EGP}$$

Recommended Configuration: Google Gemini 1.5 Flash at 32 EGP/month (within budget).

5 Total AI Services Cost Summary

5.1 Monthly Cost by Service

Service	Model	Monthly USD	Monthly EGP	Annual EGP
Vision AI	Gemini 1.5 Pro	13.13	663	7,956
Fraud Detection	Gemini 1.5 Flash	1.35	68	816
Impact Economist	GPT-4o	31.50	1,591	19,092
Graph Synthesis	GPT-4o	15.75	795	9,540
Chat Assistant	Gemma 4 31B IT	0.94	47	564
Verification Orchestrator	Gemma 4 31B IT	1.51	76	912
Report Generator	Gemini 1.5 Flash	0.63	32	384
TOTAL (Recommended)		**64.81**	**3,272**	**39,264**

Table 15: Total Monthly AI Services Cost – Recommended Configuration

5.2 Monthly Cost by Provider

Provider	Monthly USD	Monthly EGP	% of Total	Services
Google (Gemini)	15.11	763	23.3%	3 services
OpenAI (GPT-4o)	47.25	2,386	72.9%	2 services
OpenRouter (Gemma)	2.45	123	3.8%	2 services
TOTAL	**64.81**	**3,272**	**100%**	**7 services**

Table 16: Cost Distribution by AI Provider

5.3 Monthly Cost by Category

Category	Monthly EGP	Annual EGP	% of Total
AI Model Inference (LLM)	3,172	38,064	96.9%
Embedding Services (Qdrant)	100	1,200	3.1%
TOTAL AI SERVICES	**3,272**	**39,264**	**100%**

Table 17: Cost by Category

6 Deployment Scenarios

Three deployment scenarios are defined based on platform scale and user volume. All scenarios operate within the 500–10,000 EGP monthly budget range per service.

6.1 Scenario Definitions

Parameter	Development	Production	Enterprise
Monthly MAU	Up to 1,000	5,000–10,000	50,000+
Daily Evidence Submissions	50–200	500–2,000	5,000–20,000
Daily Chat Messages	100–500	1,000–5,000	10,000–50,000
Daily Report Requests	5–20	50–200	500–2,000
AI Services Active	Vision, Fraud	All 7 services	All 7 + fallback
Primary Provider	Google only	Google + OpenAI	All 3 providers
Fallback Strategy	None	OpenAI GPT-4o	Open-source models

Table 18: Deployment Scenario Parameters

6.2 Scenario Cost Comparison

Service	Development (EGP)	Production (EGP)	Enterprise (EGP)
Vision AI	663	663	663
Fraud Detection	68	68	68
Impact Economist	—	1,591	1,591
Graph Synthesis	—	795	795
Chat Assistant	—	47	47
Verification Orchestrator	—	76	76
Report Generator	—	32	32
Embedding Services	100	100	100
Volume Discount (25%)	—	—	$-1,233$
Monthly Total	**831**	**3,372**	**2,139**
Annual Total	**9,972**	**40,464**	**25,668**
Per-User/Month	**0.83**	**0.34**	**0.04**

Table 19: Deployment Scenario Cost Comparison

Note: Enterprise scenario achieves lower total cost despite serving 50x more users due to:

- Volume discount agreements with providers (25% reduction on OpenAI)
- Prompt caching reducing input token costs by 50–75%
- Model fallback to cheaper alternatives for non-critical tasks
- Embedding cost amortized across larger user base

7 Statistical Analysis

7.1 Token Usage Statistics

Metric	Development	Production	Enterprise
Total Daily Input Tokens	330,000	740,000	3,700,000
Total Daily Output Tokens	290,000	490,000	2,450,000
Total Daily Tokens	620,000	1,230,000	6,150,000
Monthly Tokens	18,600,000	36,900,000	184,500,000
Monthly API Calls (est.)	600	3,000	30,000
Average Tokens/Call	1,033	410	205

Table 20: Token Usage Statistics by Scenario

7.2 Cost Efficiency Metrics

Metric	Development	Production	Enterprise
Cost per 1K Tokens (EGP)	0.18	0.09	0.02
Cost per API Call (EGP)	1.39	1.12	0.07
Cost per Evidence (EGP)	4.16	1.69	0.17
Cost per Report (EGP)	83.10	11.24	1.07
Cost per User/Month (EGP)	0.83	0.34	0.04
Monthly Savings at 50% Cache	0	412	1,233

Table 21: Cost Efficiency Metrics by Scenario

7.3 Growth Projections

Month	MAU	Daily Tokens	Monthly EGP	Revenue EGP	Margin
Month 1–3	500	310,000	1,636	2,100	+28%
Month 4–6	2,000	620,000	3,272	4,200	+28%
Month 7–9	5,000	1,230,000	3,372	9,500	+182%
Month 10–12	10,000	2,460,000	5,200	19,000	+265%
Year 2	25,000+	4,920,000	7,800	47,500	+509%

Table 22: 12-Month Growth Projection

Assumptions:

- User growth rate: 15% month-over-month (conservative)
- Token usage scales linearly with MAU
- Per-user cost: 4.24 EGP/month (Development), 1.91 EGP/month (Production), 0.42 EGP/-month (Enterprise)
- Revenue calculated at user-charged rates (see Section 10)

8 Financial Analysis

8.1 Cost Breakdown Analysis

Cost Component	Monthly EGP	% of Budget
Google Gemini Inference	763	23.3%
OpenAI GPT-4o Inference	2,386	72.9%
OpenRouter Gemma Inference	123	3.8%
Total AI Inference	**3,272**	**100%**

Table 23: Cost Breakdown by Provider

8.2 Return on Investment (ROI)

Metric	Value
Monthly AI Cost (Production)	3,372 EGP
Annual AI Cost (Production)	40,464 EGP
Break-Even Point (Production)	3,372 users paying 1.91 EGP/month
Break-Even Point (Development)	831 users paying 4.24 EGP/month
Break-Even Point (Enterprise)	2,139 users paying 0.42 EGP/month
Cost Recovery Target	100% from user fees
Gross Margin at Production Scale	28%
Gross Margin at Enterprise Scale	265%
Payback Period	3 months (at Development scale)

Table 24: ROI and Break-Even Analysis

8.3 Sensitivity Analysis

Scenario	+20% Usage	+50% Usage	Double Usage	Provider Change
Vision AI	796 EGP	995 EGP	1,326 EGP	663→1,326
Fraud Detection	82 EGP	102 EGP	136 EGP	68→2,273
Impact Economist	1,909 EGP	2,387 EGP	3,182 EGP	1,591→2,176
Graph Synthesis	954 EGP	1,193 EGP	1,590 EGP	795→1,088
Total	3,927 EGP	4,867 EGP	6,541 EGP	—
Within Budget?	**YES**	**YES**	**YES**	Varies

Table 25: Sensitivity Analysis – Usage and Provider Variations

Key Finding: Even with 2x usage, total AI cost remains under 10,000 EGP/month with the recommended Google-primary configuration. Switching Impact Economist from GPT-4o to Gemini 1.5 Pro reduces cost by 465 EGP/month.

9 Admin Control Panel for AI Service Usage

The MASAREN platform includes an administrative interface for controlling AI service usage per user to manage costs effectively.

9.1 Usage Quota Management

Quota Parameter	Development	Production	Enterprise
Daily Token Limit (per user)	10,000	50,000	200,000
Daily API Call Limit (per user)	50	200	1,000
Monthly Token Cap (per user)	200,000	1,000,000	5,000,000
Overage Rate	2x base	1.5x base	1.2x base
Quota Reset	Account anniversary	Account anniversary	Account anniversary

Table 26: Usage Quota Configuration by Tier

9.2 Model Selection and Routing

1. **Primary Models:** Google Gemini 1.5 Pro/Flash (lowest cost)
2. **Fallback Models:** OpenAI GPT-4o/GPT-4o mini (highest quality)
3. **Budget Models:** Gemma 4 31B IT via OpenRouter (Apache 2.0, lowest cost)
4. **Automatic Routing:** Gateway routes to cheapest model meeting quality threshold
5. **Per-User Model Affinity:** Based on user tier and use case

9.3 Cost Monitoring Thresholds

Trigger	Threshold	Action
Per-User Cost	Exceeds 2x tier average	Admin notification
Daily Token Usage	Exceeds 80% quota	Automatic user warning
Monthly Burn Rate	Exceeds budget +15%	Alert email to admin
Quota Utilization	100% reached	API calls blocked, overage charged
Provider Latency	Exceeds 10s p95	Automatic failover to fallback
Error Rate	Exceeds 5%	Admin alert, model switch

Table 27: Cost Monitoring and Alert Thresholds

10 User-Charged Financial Plan

The MASAREN platform implements a user-charged AI services model where AI service costs are billed directly to platform users rather than being absorbed by the platform operator. This ensures sustainable infrastructure funding while keeping platform costs predictable.

10.1 Objective

1. Recover 100% of AI service costs from platform users
2. Maintain predictable cost structure for platform planning
3. Provide administrators tools to control and optimize AI usage
4. Ensure transparent billing for end users

10.2 Per-User Cost Schedule

Tier	Users	Daily Tokens	Cost/User (EGP/month)	Included Services
Development	Up to 1,000	350,000	4.24	Vision AI + Fraud Detection
Production	5,000+	850,000	1.91	All 7 AI services
Enterprise	50,000+	2,000,000	0.42	All AI + volume discounts

Table 28: Per-User Cost Schedule (500–10,000 EGP/month range)

10.3 Revenue Projections

Scenario	Users	Cost/User (EGP)	Monthly Revenue	Cost Recovery
Development	1,000	4.24	4,240 EGP	510%
Production	5,000	1.91	9,550 EGP	283%
Enterprise	50,000	0.42	21,000 EGP	985%

Table 29: Revenue Projections per User Tier

10.4 Billing and Payment Integration

1. **Monthly Invoice:** AI costs itemized separately on monthly user invoice
2. **Automatic Payment:** Linked to user's existing payment method on file
3. **Usage Transparency:** Users can view their token usage and costs in dashboard
4. **Quota Management:** Users can adjust their AI service usage quota via account settings
5. **Overage Notifications:** Users notified when approaching/exceeding quotas

10.5 Cost Recovery Guarantee

Guarantee	Status
Cost Recovery Rate	100% (all AI costs billed to users)
Minimum Revenue Guarantee	3,272 EGP/month (at Production scale)
Maximum Cost Exposure	10,000 EGP/month (budget cap per service)
Platform Subsidy	0 EGP (zero platform cost for AI)

Table 30: Cost Recovery Guarantees

11 Financial Risk Assessment

Risk Category	Probability	Impact	Mitigation Strategy
AI Cost Overruns	Medium	High	Usage quotas, prompt optimization, model caching
Provider Price Increases	Low	Medium	Multi-provider strategy, negotiate annual commitments
API Rate Limits	Medium	Medium	Request queuing, batch processing, fallback models
User Growth Faster Than Expected	Medium	High	Scale AI infrastructure proactively, volume discounts
Model Deprecation	Low	Medium	Maintain multi-provider support, version pinning
Currency Fluctuation (EGP/USD)	High	Medium	Fix pricing in EGP, quarterly rate reviews
Token Usage Spike	Medium	Medium	Rate limiting, auto-scaling quotas, burst budgets

Table 31: Financial Risk Assessment Matrix

11.1 Risk Mitigation Cost Budget

Mitigation Measure	Monthly Cost	Savings Potential
Prompt Caching (30–50% input reduction)	0 EGP	385–642 EGP
Model Fallback to Gemma/Gemini Flash	0 EGP	1,200–2,400 EGP
Batch Processing (off-peak)	0 EGP	100–300 EGP
Token Quota Enforcement	0 EGP	Variable

Table 32: Cost Mitigation Measures (No Additional Cost)

12 Implementation Roadmap

12.1 Phase 1: Immediate (Months 0–3)

1. Deploy per-user quota system with default limits
2. Implement prompt caching for 30–50% cost reduction
3. Configure model routing: Google primary, OpenAI fallback
4. Enable cost monitoring and admin alerts
5. Bill AI costs alongside platform subscription fees
6. Target: 90% cost recovery from first 1,000 users
7. Budget: 831 EGP/month (Development scenario)

12.2 Phase 2: Growth (Months 3–12)

1. Activate all 7 AI services (add Economist, Graph, Chat, Verification, Reports)
2. Implement model fallback system for 60–80% cost reduction
3. Deploy custom pricing tiers per user category
4. Integrate volume discount automatic application
5. Negotiate annual commitments with Google and OpenAI
6. Target: 95–100% cost recovery at scale
7. Budget: 3,372 EGP/month (Production scenario)

12.3 Phase 3: Maturity (Year 2+)

1. Negotiate enterprise pricing with OpenAI/Google
2. Deploy custom fine-tuned open-source models for high-volume tasks
3. Implement advanced predictive usage management
4. Cross-subsidize development phase from enterprise revenues
5. Target: Profitable AI services operation
6. Budget: 2,139 EGP/month (Enterprise scenario with discounts)

13 Business Terms Glossary (Financial Audit Reference)

Term	Definition
MAU (Monthly Active Users)	Unique users who perform at least one action per calendar month
Token	Minimum unit of text processed by an LLM; approximately 4 characters of English text
Input Tokens	Tokens sent to the LLM as part of a prompt (user message, system instructions, context)
Output Tokens	Tokens generated by the LLM as a response
Cost per Token (EGP)	Monthly AI cost divided by total monthly tokens processed
Cost per User/Month (EGP)	Monthly AI cost divided by number of active users
Prompt Caching	Technique to store and reuse frequently sent prompt prefixes, reducing input token costs by 50–75%
Model Fallback	Automatic switching to a cheaper or more available model when the primary model is unavailable or rate-limited
Rate Limiting	Restriction on the number of API calls per time window to prevent cost overruns
Quota	Maximum allocated token/call limit per user per time period
Overage	Usage exceeding the allocated quota, charged at a premium rate
User-Charged Model	Billing structure where AI costs are passed to end users rather than absorbed by the platform operator
Cost Recovery Rate	Percentage of AI service costs recovered through user fees; target is 100%
Gross Margin	Revenue minus AI costs, expressed as a percentage of revenue
Break-Even Point	Number of paying users required for AI cost revenue to equal AI service expenses
Volume Discount	Pricing reduction offered by AI providers for high-volume usage commitments
Embedding	Numerical vector representation of text for semantic search; generated by embedding models
Latency (p50/p95/p99)	Response time percentiles; p95 = 95% of requests complete within this time
CrewAI	Multi-agent AI framework used for complex verification workflows
LLM (Large Language Model)	Foundation model capable of text generation, analysis, and reasoning
EGP	Egyptian Pound; primary currency for all financial projections in this report

Table 33: Business Terms Glossary for Financial Audit

This report was prepared by the MASAREN Financial Planning Division.
All pricing reflects August 2026 market rates.
This document is intended for financial audit and business planning purposes.
Exchange rate: 1 USD = 50.5 EGP.
Budget range: 500–10,000 EGP/month per AI service.